Homework 3

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Problem 1

Let $(F, +, \cdot)$ be a field.

(a) Prove that the multiplicative identity is unique: If $b \in F$ satisfies $b \cdot a = a \cdot b = a$ for all $a \in F$, then $b = 1$.

(b) Prove that multiplicative inverses are unique: If $a \in F \setminus \{0\}$ and $b \in F$ satisfy $b \cdot a = a \cdot b = 1$, then $b = a^{-1}$.

(c) Prove that if $a \cdot b = a \cdot c$ and $a \neq 0$, then $b = c$.

(d) Prove that if $a, b \in F$ then $-a + (-b) = -(a + b)$.

(e) Prove that if $a, b \in F$ and $a, b \neq 0$, then $a^{-1} \cdot b^{-1} = (ab)^{-1}$.

Discussion & Scratch Work

Main ideas (and why we cannot "just divide"). In a general field, "division" is not a primitive operation. Instead, we justify any division-like step by introducing multiplicative inverses via **(M5)** and then using associativity **(M3)** and the identity **(M4)**.

- **(a)** Compare $b \cdot 1$ computed from the hypothesis and from **(M4)**.
- **(b)** Insert $1 = a \cdot a^{-1}$ (from **(M5)**) and reassociate (by **(M3)**) to force $b = a^{-1}$.
- **(c)** Prove cancellation by multiplying $a \cdot b = a \cdot c$ on the left by a^{-1} (existence from **(M5)**), then use **(M3)** and **(M4)**.
- **(d)** Show $(-a) + (-b)$ is an additive inverse of $(a + b)$ by checking $(a + b) + ((-a) + (-b)) = 0$ using **(A2)**, **(A3)**, **(A4)**, **(A5)**. Conclude it equals $-(a + b)$ by uniqueness of additive inverses (proved from the axioms, as in lecture).
- **(e)** Show $a^{-1} \cdot b^{-1}$ is an inverse of ab using commutativity **(M2)**, associativity **(M3)**, and **(M5)**. Then apply (b) to conclude $a^{-1}b^{-1} = (ab)^{-1}$.

Proof.
Let $(F, +, \cdot)$ be a field.

(a) Suppose $b \in F$ satisfies

$$b \cdot a = a \cdot b = a \quad \text{for all } a \in F.$$

Let $1 \in F$ be the multiplicative identity whose existence is guaranteed by **(M4)**. Setting $a = 1$ in the hypothesis gives

$$b \cdot 1 = 1.$$

But by **(M4)** (applied with $a = b$), we also have $b \cdot 1 = b$. Therefore $b = 1$, so the multiplicative identity is unique.

(b) Suppose $a \in F \setminus \{0\}$ and $b \in F$ satisfy

$$b \cdot a = a \cdot b = 1.$$

By the multiplicative inverse axiom **(M5)**, there exists $a^{-1} \in F$ such that

$$a \cdot a^{-1} = a^{-1} \cdot a = 1.$$

We prove that $b = a^{-1}$. Using the multiplicative identity axiom **(M4)**, we have

$$b = b \cdot 1.$$

Substituting $1 = a \cdot a^{-1}$ from **(M5)** gives

$$b = b \cdot (a \cdot a^{-1}).$$

By associativity of multiplication **(M3)**,

$$b \cdot (a \cdot a^{-1}) = (b \cdot a) \cdot a^{-1}.$$

Using the hypothesis $b \cdot a = 1$, this becomes

$$(b \cdot a) \cdot a^{-1} = 1 \cdot a^{-1}.$$

Finally, by **(M4)** we have $1 \cdot a^{-1} = a^{-1}$. Therefore $b = a^{-1}$, so the multiplicative inverse of a is unique.

(c) Suppose $a, b, c \in F$ satisfy $a \cdot b = a \cdot c$ and $a \neq 0$. By **(M5)**, there exists $a^{-1} \in F$ such that $a^{-1} \cdot a = 1$. Multiply the equality $a \cdot b = a \cdot c$ on the left by a^{-1}:

$$a^{-1} \cdot (a \cdot b) = a^{-1} \cdot (a \cdot c).$$

By associativity **(M3)**,

$$(a^{-1} \cdot a) \cdot b = (a^{-1} \cdot a) \cdot c.$$

Using $a^{-1} \cdot a = 1$ (from **(M5)**) and the identity axiom **(M4)**, we get

$$1 \cdot b = 1 \cdot c \quad \Rightarrow \quad b = c.$$

(d) We claim that $-a + (-b)$ is the additive inverse of $a + b$, hence equals $-(a + b)$. First compute, using associativity **(A3)** and commutativity **(A2)** of $+$,

$$\begin{aligned}
(a+b) + \big((-a) + (-b)\big) &= a + \Big(b + \big((-a) + (-b)\big)\Big) \quad \text{by } \textbf{(A3)} \\
&= a + \Big(\big(b + (-b)\big) + (-a)\Big) \quad \text{by } \textbf{(A3)} \text{ and } \textbf{(A2)} \\
&= a + \big(0 + (-a)\big) \quad \text{by } \textbf{(A5)} \\
&= a + (-a) \quad \text{by } \textbf{(A4)} \\
&= 0 \quad \text{by } \textbf{(A5)}.
\end{aligned}$$

Thus $(-a) + (-b)$ is an additive inverse of $a + b$.

Uniqueness of additive inverses (from axioms). If $x, y \in F$ both satisfy $(a + b) + x = 0$ and $(a+b)+y = 0$, then adding $-(a+b)$ to both sides and using **(A3)**, **(A4)**, **(A5)** yields $x = y$ (as in lecture). Therefore the additive inverse of $a + b$ is unique, and since $-(a + b)$ denotes that unique inverse, we conclude

$$-a + (-b) = (-a) + (-b) = -(a + b).$$

(e) Assume $a, b \in F$ with $a \neq 0$ and $b \neq 0$. By **(M5)**, there exist $a^{-1}, b^{-1} \in F$ such that $a \cdot a^{-1} = 1$ and $b \cdot b^{-1} = 1$. Consider

$$(ab) \cdot (a^{-1} \cdot b^{-1}).$$

By associativity **(M3)** and commutativity **(M2)**,

$$(ab) \cdot (a^{-1}b^{-1}) = a \cdot b \cdot a^{-1} \cdot b^{-1} = (a \cdot a^{-1}) \cdot (b \cdot b^{-1}).$$

Using **(M5)** and then **(M4)**, this equals $1 \cdot 1 = 1$. Similarly,

$$(a^{-1}b^{-1}) \cdot (ab) = 1.$$

Hence $a^{-1}b^{-1}$ is a multiplicative inverse of ab. By part **(b)** (uniqueness of inverses),

$$a^{-1} \cdot b^{-1} = (ab)^{-1}.$$

$\square$

Problem 2

Given two points (x_1, y_1) and (x_2, y_2) in $\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}$, we write

$$(x_1, y_1) < (x_2, y_2) \Leftrightarrow x_1 < x_2 \text{ or } (x_1 = x_2 \text{ and } y_1 < y_2).$$

(a) Prove that $<$ is an order relation on $\mathbb{R}^2$.

(b) Prove that $(\mathbb{R}^2, <)$ does not satisfy the least upper bound property. (Hint: Consider the set of all points on the y-axis.)

Discussion & Scratch Work

Main ideas. This is the *lexicographic order* on $\mathbb{R}^2$.

- **(a)** To prove $<$ is an order relation on $\mathbb{R}^2$, verify the two defining properties: **trichotomy** and **transitivity**. The proofs reduce to the corresponding properties of the usual order on $\mathbb{R}$ by splitting into cases on the first coordinates.
- **(b)** To show the least upper bound property fails, use the set $A = \{(0, y) : y \in \mathbb{R}\}$ (the y-axis). It is bounded above (e.g. by $(1, 0)$), but any upper bound must have first coordinate > 0, and then you can always find a strictly smaller upper bound by decreasing the first coordinate (e.g. replace (x, y) by $(x/2, 0)$). Hence no least upper bound exists.

Proof.
Let $p = (x_1, y_1)$, $q = (x_2, y_2)$, and $r = (x_3, y_3)$ be points in $\mathbb{R}^2$.

(a) We prove that $<$ is an order relation on $\mathbb{R}^2$ in the sense of the lecture definition.

Trichotomy. By trichotomy of the usual order on $\mathbb{R}$, exactly one of $x_1 < x_2$, $x_1 = x_2$, or $x_2 < x_1$ holds.

- If $x_1 < x_2$, then $p < q$ by definition.
- If $x_2 < x_1$, then $q < p$ by definition.
- If $x_1 = x_2$, then by trichotomy on $\mathbb{R}$ applied to y_1, y_2, exactly one of $y_1 < y_2$, $y_1 = y_2$, or $y_2 < y_1$ holds. In the first case, $p < q$; in the second, $p = q$; in the third, $q < p$.

Therefore exactly one of $p < q$, $p = q$, or $q < p$ holds.

Transitivity. Assume $p < q$ and $q < r$. We show $p < r$.

If $x_1 < x_2$, then from $q < r$ we have either $x_2 < x_3$ or $x_2 = x_3$. In both cases $x_2 \leq x_3$, hence $x_1 < x_3$, so $p < r$ by definition.

If instead $x_1 = x_2$, then $p < q$ forces $y_1 < y_2$. Now consider $q < r$:

- If $x_2 < x_3$, then $x_1 = x_2 < x_3$, so $p < r$.

- If $x_2 = x_3$, then $q < r$ forces $y_2 < y_3$, and by transitivity in $\mathbb{R}$ we get $y_1 < y_3$, so (since $x_1 = x_3$) we conclude $p < r$.

Thus $<$ is transitive.

Since $<$ satisfies trichotomy and transitivity, it is an order relation on $\mathbb{R}^2$.

(b) Let

$$A := \{(0, y) : y \in \mathbb{R}\} \subseteq \mathbb{R}^2.$$

Then A is nonempty (for example $(0, 0) \in A$). We claim A is bounded above in $(\mathbb{R}^2, <)$. Indeed, $(1, 0)$ is an upper bound, because for any $(0, y) \in A$ we have $0 < 1$, hence $(0, y) < (1, 0)$ by the definition of $<$.

We show A has *no* least upper bound. First note that if (u, v) is any upper bound of A, then necessarily $u > 0$. If $u < 0$, then $(0, 0) \in A$ would satisfy $(u, v) < (0, 0)$, so (u, v) could not be an upper bound. If $u = 0$, then being an upper bound would require $y \leq v$ for all $y \in \mathbb{R}$, which is impossible.

Now take any upper bound (u, v) of A; then $u > 0$. Define $w := (u/2, 0)$. For any $(0, y) \in A$ we have $0 < u/2$, hence $(0, y) < (u/2, 0) = w$, so w is also an upper bound of A. But $w < (u, v)$ because $u/2 < u$.

Therefore every upper bound of A is strictly larger than some other upper bound of A, so no least upper bound exists. Hence $(\mathbb{R}^2, <)$ does not satisfy the least upper bound property. $\square$

Problem 3

Recall that the set of complex numbers $\mathbb{C}$ consists of all numbers of the form $a + ib$ for $a, b \in \mathbb{R}$. We can add and multiply complex numbers as follows:

$$(a + ib) + (c + id) = (a + c) + i(b + d) \quad \text{and} \quad (a + ib) \cdot (c + id) = (ac - bd) + i(ad + bc).$$

With these operations, it turns out $(\mathbb{C}, +, \cdot)$ is a field (but you do not need to prove this).

(a) The definition of a field says that for any $a + ib \neq 0 + i0$, there exists $c + id \in \mathbb{C}$ so that $(a + ib) \cdot (c + id) = 1 + i0$. What are $c, d \in \mathbb{R}$ in terms of $a, b \in \mathbb{R}$? Prove your answer.

(b) Prove that there is no order relation $<$ on $\mathbb{C}$ that makes $(\mathbb{C}, +, \cdot, <)$ an ordered field. (Hint: Can $i = 0 + i1$ be positive? Negative? Zero?)

Discussion & Scratch Work

Main ideas.

- **(a)** To find $(a+ib)^{-1}$, write an unknown inverse as $c+id$ and expand $(a+ib)(c+id) = (ac - bd) + i(ad + bc)$. Matching real and imaginary parts with $1 + i0$ gives a 2×2 linear system. Solve it to get

$$c = \frac{a}{a^2 + b^2}, \qquad d = -\frac{b}{a^2 + b^2}.$$

  The key point is that $a^2 + b^2 \neq 0$ whenever $a + ib \neq 0$.

- **(b)** In an ordered field, we always have $1 > 0$ and $x \neq 0 \Rightarrow x^2 > 0$ (Appendix, **(P3)**). But in $\mathbb{C}$ we have $i^2 = -1$. Trichotomy forces $i > 0$, $i = 0$, or $i < 0$; each case contradicts $x^2 > 0$ together with $1 > 0$. So no order can make $\mathbb{C}$ into an ordered field.

Proof.
(a) Let $a + ib \in \mathbb{C}$ with $a + ib \neq 0 + i0$. We seek $c + id \in \mathbb{C}$ such that

$$(a + ib)(c + id) = 1 + i0.$$

Using the given multiplication rule,

$$(a + ib)(c + id) = (ac - bd) + i(ad + bc).$$

Thus we require

$$ac - bd = 1, \qquad ad + bc = 0. \tag{1}$$

Multiply the first equation in (1) by a and the second by b, then add:

$$a(ac - bd) + b(ad + bc) = a.$$

The left-hand side simplifies to $(a^2 + b^2)c$, so

$$(a^2 + b^2)c = a.$$

Similarly, multiply the second equation by a and the first by b, then subtract:

$$a(ad + bc) - b(ac - bd) = 0 - b,$$

whose left-hand side simplifies to $(a^2 + b^2)d$, so

$$(a^2 + b^2)d = -b.$$

Because $a + ib \neq 0$, we have $(a, b) \neq (0, 0)$, hence $a^2 + b^2 > 0$ and in particular $a^2 + b^2 \neq 0$. Therefore we may divide by $a^2 + b^2$ to obtain

$$c = \frac{a}{a^2 + b^2}, \qquad d = -\frac{b}{a^2 + b^2}.$$

A direct substitution into (1) (or into $(ac - bd) + i(ad + bc)$) shows that these values indeed yield $(a + ib)(c + id) = 1 + i0$. This proves the claimed formula for the multiplicative inverse.

(b) Suppose, for contradiction, that there exists an order relation $<$ on $\mathbb{C}$ such that $(\mathbb{C}, +, \cdot, <)$ is an ordered field. By Appendix **(P3)**, we have $1 > 0$ and, for any $x \in \mathbb{C}$, $x \neq 0 \Rightarrow x^2 > 0$. By trichotomy, exactly one of $i > 0$, $i = 0$, or $i < 0$ holds. We treat each case:

- If $i = 0$, then $i^2 = 0$, contradicting $i^2 = -1$.
- If $i > 0$, then $i \neq 0$, so Appendix **(P3)** implies $i^2 > 0$. But $i^2 = -1$, so $-1 > 0$, hence $1 < 0$, contradicting $1 > 0$.
- If $i < 0$, then $-i > 0$ (using Appendix **(P1)** with $a = i$). Since $-i \neq 0$, Appendix **(P3)** gives $(-i)^2 > 0$. But $(-i)^2 = i^2 = -1$, leading again to $-1 > 0$ and hence $1 < 0$, a contradiction.

All cases contradict the ordered-field consequences, so no such order relation exists. $\square$

Problem 4

Let $A \subseteq \mathbb{R}$ be a nonempty and bounded set, let $c \in \mathbb{R}$ be a number, and define

$$B = \{c \cdot a : a \in A\}.$$

(a) Prove that if $c > 0$, then $\sup B = c \cdot \sup A$.

(b) Prove that if $c < 0$, then $\sup B = c \cdot \inf A$.

Discussion & Scratch Work

Proof.

$\square$

Problem 5

Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded sets, and define

$$S = \{a + b : a \in A \text{ and } b \in B\}.$$

Prove that $\sup S = \sup A + \sup B$.

Discussion & Scratch Work

Proof.

$\square$

Problem 6

Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded sets. Prove or disprove the following statements:

(a) If $a \leq b$ for all $a \in A$ and for all $b \in B$, then $\sup A \leq \inf B$.

(b) If $a < b$ for all $a \in A$ and for all $b \in B$, then $\sup A < \inf B$.

Discussion & Scratch Work

Proof.

$\square$

Appendix: Toolbox

Field axioms (as used in lecture)

(A1) **(Closure for $+$)** If $a, b \in F$, then $a + b \in F$.

(A2) **(Commutativity of $+$)** If $a, b \in F$, then $a + b = b + a$.

(A3) **(Associativity of $+$)** If $a, b, c \in F$, then $(a + b) + c = a + (b + c)$.

(A4) **(Additive identity)** There exists $0 \in F$ such that $a + 0 = 0 + a = a$ for all $a \in F$.

(A5) **(Additive inverses)** For each $a \in F$ there exists $-a \in F$ such that $a + (-a) = (-a) + a = 0$.

(M1) **(Closure for $\cdot$)** If $a, b \in F$, then $a \cdot b \in F$.

(M2) **(Commutativity of $\cdot$)** If $a, b \in F$, then $a \cdot b = b \cdot a$.

(M3) **(Associativity of $\cdot$)** If $a, b, c \in F$, then $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.

(M4) **(Multiplicative identity)** There exists $1 \in F$ such that $a \cdot 1 = 1 \cdot a = a$ for all $a \in F$.

(M5) **(Multiplicative inverses)** For each $a \in F \setminus \{0\}$ there exists $a^{-1} \in F$ such that $a \cdot a^{-1} = a^{-1} \cdot a = 1$.

(D) **(Distributivity)** If $a, b, c \in F$, then $(a + b) \cdot c = a \cdot c + b \cdot c$.

Derived properties of fields (proved from the axioms in Lecture 7)

(1) The additive and multiplicative identities are unique.

(2) Additive and multiplicative inverses are unique.

(3) If $a, b, c \in F$ satisfy $a + b = a + c$, then $b = c$. If $a, b, c \in F$ satisfy $a \cdot b = a \cdot c$ and $a \neq 0$, then $b = c$.

(4) For all $a \in F$, we have $a \cdot 0 = 0 \cdot a = 0$.

(5) For all $a, b \in F$, we have $(-a) \cdot b = a \cdot (-b) = -(a \cdot b)$.

(6) For all $a, b \in F$, we have $(-a) \cdot (-b) = a \cdot b$.

(7) If $a \cdot b = 0$, then $a = 0$ or $b = 0$.

Bounds, maxima/minima, and suprema/infima (Lecture 8)

Bounds and extrema

Definition 1. *Let $(F, +, \cdot, <)$ be an ordered field, and let $A \subseteq F$ be nonempty.*

(B1) *We say that A is* bounded above *if there exists $M \in F$ such that $a \leq M$ for all $a \in A$. Any such M is called an* upper bound *for A. If, in addition, $M \in A$, then M is called the* maximum *of A.*

(B2) *We say that A is* bounded below *if there exists $m \in F$ such that $m \leq a$ for all $a \in A$. Any such m is called an* lower bound *for A. If, in addition, $m \in A$, then m is called the* minimum *of A.*

(B3) *We say that A is* bounded *if it is both bounded above and bounded below.*

Remark. *Some sets do not have a maximum (or a minimum). If a maximum exists, then it is unique: if $M, M' \in A$ both satisfy $a \leq M$ for all $a \in A$ and $a \leq M'$ for all $a \in A$, then $M \leq M'$ and $M' \leq M$, hence $M = M'$. (The same statement holds for minima.)*

Remark. *Even if a set does not have a maximum, it may still have (at least one) upper bound.*

Least upper bounds and greatest lower bounds

Definition 2. *Let $(F, +, \cdot, <)$ be an ordered field, and let $A \subseteq F$ be nonempty.*

(S1) *A number $L \in F$ is called the* least upper bound *(or* supremum*) of A if:*

- *L is an upper bound for A, and*
- *if M is an upper bound for A, then $L \leq M$.*

When this is true, we write $L = \sup A$.

(S2) *A number $\ell \in F$ is called the* greatest lower bound *(or* infimum*) of A if:*

- *ℓ is a lower bound for A, and*
- *if m is a lower bound for A, then $m \leq \ell$.*

When this is true, we write $\ell = \inf A$.

Lemma. *The least upper bound of a set is unique, if it exists.*

Proof. Suppose $A \subseteq F$ and $L, M \in F$ are both least upper bounds for A. Because L is a least upper bound, it is an upper bound; because M is a least upper bound, it is an upper bound. Since L is the *least* upper bound and M is an upper bound, we have $L \leq M$. Similarly, since M is the *least* upper bound and L is an upper bound, we have $M \leq L$. Therefore $L = M$. $\square$

Order relations and ordered fields (Lectures 7–8)

Order relations

Definition 3. *Let A be a nonempty set. An* order relation *(or* inequality*) on A is a relation $<$ on A such that:*

(OR1) ***(Trichotomy)*** *For all $a, b \in A$, exactly one of the following holds: $a < b$, or $a = b$, or $b < a$.*

(OR2) ***(Transitivity)*** *For all $a, b, c \in A$, if $a < b$ and $b < c$, then $a < c$.*

Notation. We write

$$a > b \text{ if } b < a, \qquad a \leq b \text{ if } (a < b) \text{ or } (a = b), \qquad a \geq b \text{ if } (b < a) \text{ or } (a = b).$$

Ordered fields

Definition 4. *Let $(F, +, \cdot)$ be a field. We say that $(F, +, \cdot, <)$ is an* ordered field *if it is equipped with an order relation $<$ on F such that:*

(O1) *If $a, b, c \in F$ satisfy $a < b$, then $a + c < b + c$.*

(O2) *If $a, b, c \in F$ satisfy $a < b$ and $0 < c$, then $a \cdot c < b \cdot c$.*

Useful consequences (proved in Lecture 8). Let $(F, +, \cdot, <)$ be an ordered field.

(P1) $a > 0 \iff -a < 0$.

(P2) If $a, b, c \in F$ satisfy $a < b$ and $c < 0$, then $a \cdot c > b \cdot c$.

(P3) If $a \in F$ satisfies $a \neq 0$, then $a^2 = a \cdot a > 0$. In particular, $1 > 0$.

(P4) If $a, b \in F$ satisfy $0 < a < b$, then $0 < b^{-1} < a^{-1}$.